

Unit –I Introduction to machine learning, scope and limitations, regression, probability, statistics and linear algebra for machine learning, convex optimization, data visualization, hypothesis function and testing, data distributions, data preprocessing, data augmentation, normalizing data sets, machine learning models, supervised and unsupervised learning.

★ Introduction to Machine Learning:

Machine Learning (ML) is a field of computer science and artificial intelligence that enables computers to learn from data and make decisions or predictions without being explicitly programmed for every task.

Instead of writing fixed rules, we provide data, and the system learns patterns from it.

What is Machine Learning?

Machine Learning is a subset of Artificial Intelligence (AI) where algorithms improve automatically through experience. The more data the system processes, the better it becomes at performing a task.

In traditional programming:

Input + Rules → Output

In Machine Learning:

Input + Output → Model (learned rules)

The system learns the rules automatically from data.

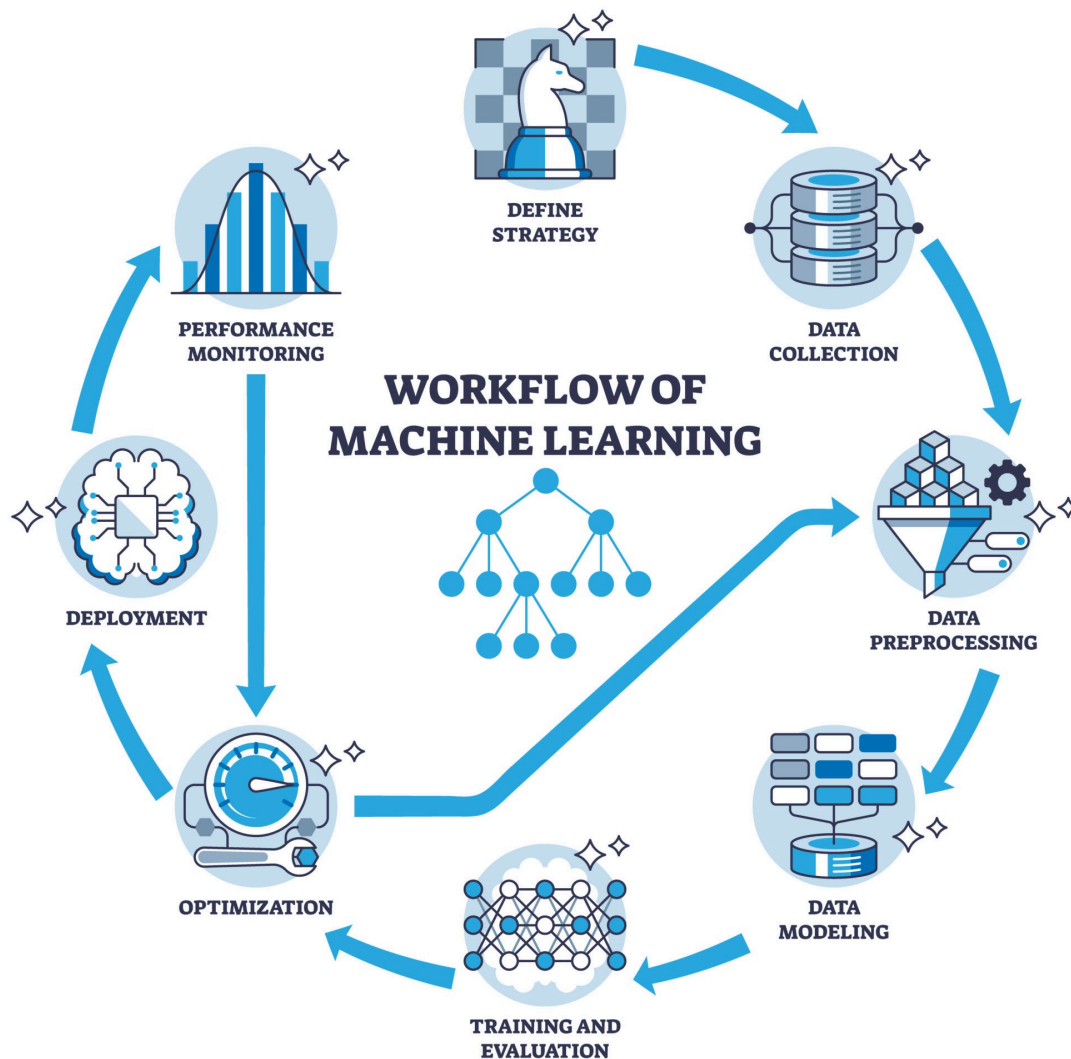
For example:

- Email spam detection
- Movie recommendations
- Voice assistants
- Fraud detection in banking

Machine Learning Works

At a high level, ML follows these steps:

1. **Collect Data** – Gather relevant data.
2. **Prepare Data** – Clean and organize the data.
3. **Choose a Model** – Select an algorithm.
4. **Train the Model** – Feed data into the algorithm.
5. **Evaluate the Model** – Test accuracy and performance.
6. **Make Predictions** – Use the trained model on new data.



Key Concepts in Machine Learning

- **Dataset** – Collection of data used for training.
- **Features** – Input variables.
- **Label/Target** – Output variable.
- **Model** – The mathematical representation learned from data.
- **Training** – Learning from data.
- **Testing** – Evaluating performance.
- **Overfitting** – Model performs well on training data but poorly on new data.
- **Underfitting** – Model is too simple to capture patterns.

Applications of Machine Learning

Machine Learning is used in:

- Healthcare (disease prediction)
- Finance (risk analysis)
- Marketing (customer targeting)
- Transportation (traffic prediction)
- Natural Language Processing (chatbots, translation)

Why Machine Learning is Important

- Handles large amounts of data
- Improves decision-making
- Automates complex tasks
- Enables intelligent systems

Scope and Limitations of Machine Learning:

Machine Learning has a broad and growing range of applications. Key areas include:

1. Data Analysis and Prediction

- ML can process large datasets and uncover patterns humans might miss.
- Examples: Stock price prediction, weather forecasting, sales forecasting.

2. Automation

- Replaces repetitive tasks and reduces human intervention.
- Examples: Automated customer support, chatbots, email filtering.

3. Personalization

- Tailors recommendations and services based on user behavior.
- Examples: Netflix recommendations, Amazon product suggestions, targeted advertising.

4. Healthcare

- Assists in diagnosis, treatment planning, and drug discovery.
- Examples: Disease prediction from medical imaging, patient risk assessment.

5. Finance

- Detects fraud, predicts credit risk, and automates trading decisions.
- Examples: Credit card fraud detection, algorithmic trading.

6. Natural Language Processing (NLP)

- Enables machines to understand and generate human language.
- Examples: Chatbots, language translation, sentiment analysis.

7. Robotics and Autonomous Systems

- Enables machines to act intelligently in dynamic environments.
- Examples: Self-driving cars, robotic process automation, drones.

8. Computer Vision

- ML models interpret visual data from images and videos.
- Examples: Face recognition, surveillance, medical imaging analysis.

9. Research and Innovation

- Helps in discovering patterns, trends, and new solutions in scientific research.
- Examples: Climate modeling, genomics, material science.

Despite its power, ML has some inherent limitations:

1. Data Dependence

- ML models require large, high-quality datasets.
- Poor or biased data leads to inaccurate predictions.

2. Computational Cost

- Training complex models (like deep learning) requires significant computing power and time.

3. Lack of Common Sense

- ML models cannot reason or understand context like humans.
- Example: A model may misclassify images if presented in unusual contexts.

4. Overfitting and Underfitting

- Overfitting: Model learns training data too well but fails on new data.
- Underfitting: Model is too simple to capture the underlying patterns.

5. Interpretability Issues

- Many advanced ML models (e.g., deep neural networks) act as “black boxes,” making it hard to explain decisions.

6. Bias and Fairness

- Models may inherit biases from training data, leading to unfair or discriminatory outcomes.

7. Security Concerns

- Vulnerable to adversarial attacks (small input changes causing wrong outputs).
- Example: Fooling a self-driving car's vision system.

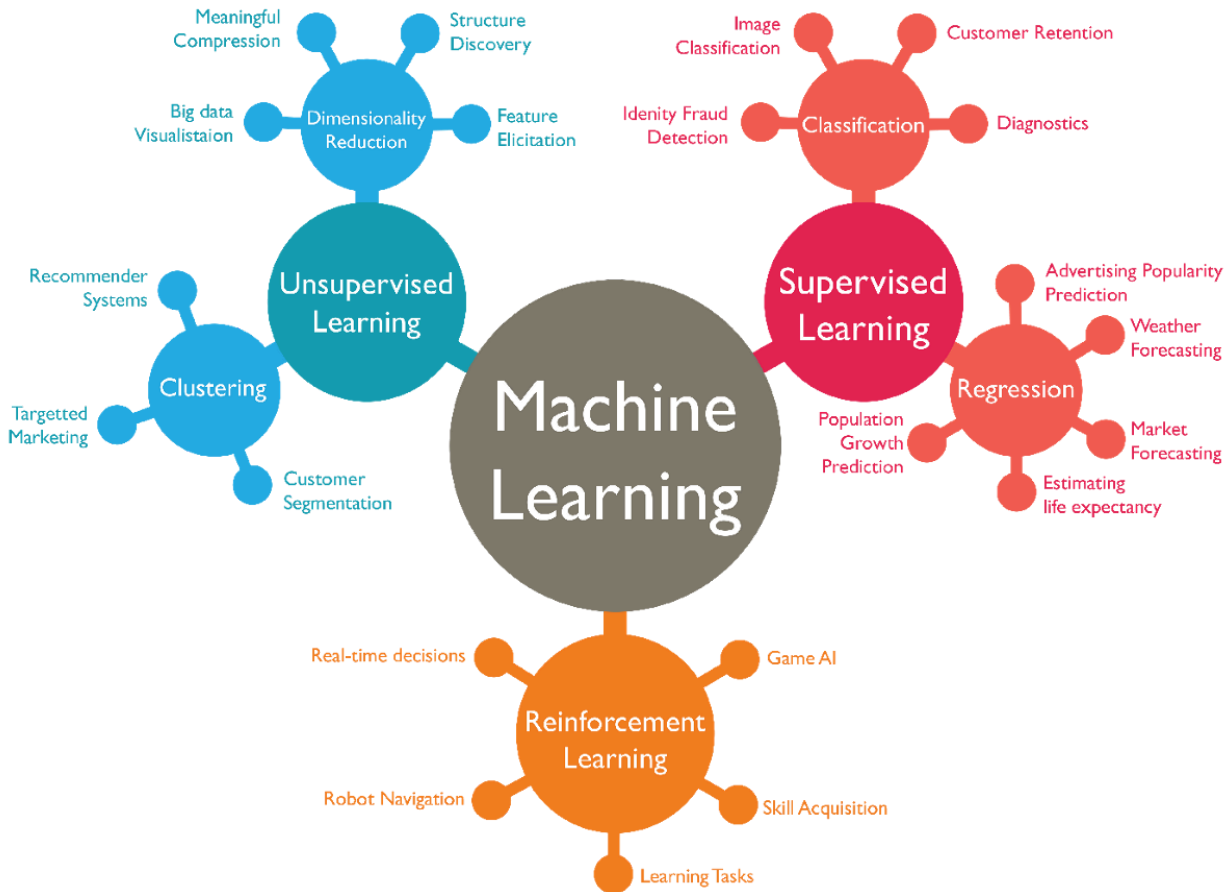
8. Limited Transferability

- Models trained for one task may fail in a different environment without retraining.

Aspect	Scope / Benefit	Limitation / Challenge
Data	Learn from large datasets	Requires high-quality, clean data
Automation	Saves human effort	Cannot replace human intuition fully
Personalization	Tailored experiences	Privacy concerns
Predictions	Forecast trends accurately	Vulnerable to bias & errors
Real-world deployment	Enables intelligent systems	Computationally expensive, security risk

Types of Machine Learning: Machine Learning (ML) is mainly divided into **4 major types**

1. **Supervised Learning**
 - A) Regression
 - B) Classification
2. **Unsupervised Learning**
 - A) Clustering
 - B) Dimensionality Reduction
3. **Semi-Supervised Learning**
4. **Reinforcement Learning**



Supervised learning is a type of Machine Learning where the model is trained using **labeled data**.

Labeled data means:

- Input data is given
- Correct output (answer) is also given

The model learns the relationship between input and output and then predicts results for new data.

How It Works

1. Provide training data (input + correct output).
2. Model learns the mapping.
3. Model predicts output for new data.

Example:

If we give: Size of house → Price (The model learns how size affects price.)

A) Regression (Output = Numeric Value) : Used when output is a number.

Examples:

- House price prediction
- Salary prediction
- Temperature forecasting
- Sales prediction
- Stock price prediction

B) Classification (Output = Category): Used when output is a category.

Examples:

- Spam / Not Spam
- Pass / Fail
- Disease Yes / No
- Fraud / Not Fraud
- Image classification (Cat/Dog)

Unsupervised learning is used when **no labeled data** is provided. The model tries to find hidden patterns, groups, or structures in data. There is no correct answer given.

How It Works

1. Provide only input data.
2. Model finds similarities and patterns.
3. Groups similar data points together.

Example:

If we give customer purchase data, it groups customers with similar buying behavior.

A) Clustering: Grouping similar data points.

Examples:

- Customer segmentation
- Grouping similar products
- Social network grouping
- Student grouping by performance

B) Association: Finding relationships between variables.

Example:

- Market Basket Analysis (If customer buys bread → also buys butter)

C) Dimensionality Reduction: Reducing number of input variables while keeping important information.

Example:

- Image compression
- Data visualization

Semi-Supervised Learning is a combination of Supervised Learning & Unsupervised Learning .In this Some data is labeled ,Most data is unlabeled.(Labeling data is expensive and time-consuming. So we use small labeled data + large unlabeled data.)

Examples

- Medical image classification
- Face recognition
- Speech recognition
- Text classification
- Fraud detection

Reinforcement Learning (RL) is based on a reward and punishment system.An agent learns by interacting with environment..The Basic Components are-

- Agent → Learner
- Environment → Where agent works
- Action → Decision taken
- Reward → Feedback

How It Works

1. Agent takes action.
2. Environment gives reward or penalty.
3. Agent learns to maximize reward.

Example:

In a game:

- Correct move → +10 points
- Wrong move → -5 points

Over time, agent learns best strategy.

Examples

- Self-driving cars
- Game playing AI (Chess, Ludo)
- Robot movement
- Traffic light control

- Recommendation systems
- Portfolio management

Feature	Supervised	Unsupervised	Semi-Supervised	Reinforcement
Labeled Data	Yes	No	Partial	No
Output Type	Prediction	Pattern	Prediction	Action
Example	Price prediction	Customer grouping	Medical images	Game AI

ML Type	Advantages	Disadvantages
Supervised	✓ High accuracy when trained properly ✓ Easy to evaluate performance (accuracy, precision, etc.) ✓ Works well for prediction problems ✓ Many proven algorithms available	✗ Requires large labeled dataset ✗ Data labeling is expensive and time-consuming ✗ Overfitting risk ✗ Not suitable when labeled data is unavailable
Unsupervised	✓ No need for labeled data ✓ Useful for discovering hidden patterns ✓ Good for data exploration ✓ Works well with large datasets	✗ Difficult to evaluate accuracy ✗ Results may not always be meaningful ✗ Interpretation can be complex ✗ May group data incorrectly
Semi-Supervised	✓ Reduces labeling cost ✓ Better accuracy than unsupervised learning ✓ Useful when labeled data is limited	✗ More complex to implement ✗ Needs careful model tuning ✗ Risk of propagating wrong labels

	✓ Improves model generalization	✗ Requires balance between labeled & unlabeled data
Reinforcement	✓ Learns optimal decision-making ✓ Works well in dynamic environments ✓ Improves through continuous learning ✓ Useful for robotics & game AI	✗ Requires high computational power ✗ Training takes long time ✗ Complex to design reward system ✗ Trial-and-error can be costly

Examples–

Supervised Learning	Unsupervised Learning	Semi-Supervised Learning	Reinforcement Learning
Email Spam Detection (Classification)	Customer Segmentation (Clustering)	Medical Image Classification (Classification)	Self-Driving Cars (Decision-Making)
Disease Prediction (Classification)	Market Basket Analysis (Association)	Face Recognition with Limited Labels (Classification)	Game Playing AI (Sequential Decision Problem)
Credit Card Fraud Detection (Classification)	News Article Grouping (Clustering)	Speech Recognition (Classification)	Robot Navigation (Control Problem)
Handwritten Digit Recognition (Classification)	Image Compression (Dimensionality Reduction)	Text Classification (Classification)	Smart Traffic Control (Optimization Problem)
Face Recognition (Classification)	Anomaly Detection (Pattern Detection)	Fraud Detection with Partial Labels (Classification)	Industrial Automation (Control & Optimization)
Student Pass/Fail Prediction (Classification)	Social Network Analysis (Clustering)	Web Page Classification (Classification)	Portfolio Management (Optimization Problem)
Loan Approval Prediction (Classification)	Topic Modeling (Pattern Discovery)	Video Classification (Classification)	Drone Navigation (Control Problem)
Sentiment Analysis (Classification)	Product Grouping (Clustering)	Object Detection (Classification)	Personalized Advertisements (Sequential Decision Problem)
Customer Churn Prediction (Classification)	DNA Pattern Analysis (Pattern Discovery)	Handwriting Recognition (Classification)	Dynamic Recommendation System (Decision-Making)
Movie Genre Classification (Classification)	Sales Pattern Analysis (Pattern Discovery)	Email Filtering (Classification)	Energy Management System (Optimization Problem)

House Price Prediction (Regression)	Student Performance Grouping (Clustering)	Document Classification (Classification)	Supply Chain Optimization (Optimization Problem)
Salary Prediction (Regression)	Website User Behavior Analysis (Clustering)	Image Labeling with Few Labels (Classification)	Chatbot Learning via Feedback (Policy Learning)
Stock Price Prediction (Regression)	Data Visualization using PCA (Dimensionality Reduction)	Voice Command Recognition (Classification)	Smart Home Automation (Control Problem)
Weather Forecasting (Regression)	Recommendation Pre-Clustering (Clustering)	Semi-Supervised Sentiment Analysis (Classification)	Video Game AI (Sequential Decision Problem)
Sales Prediction (Regression)	Customer Buying Pattern Analysis (Association)	Medical Diagnosis with Limited Data (Classification)	Warehouse Robot Optimization (Control & Optimization)

Regression

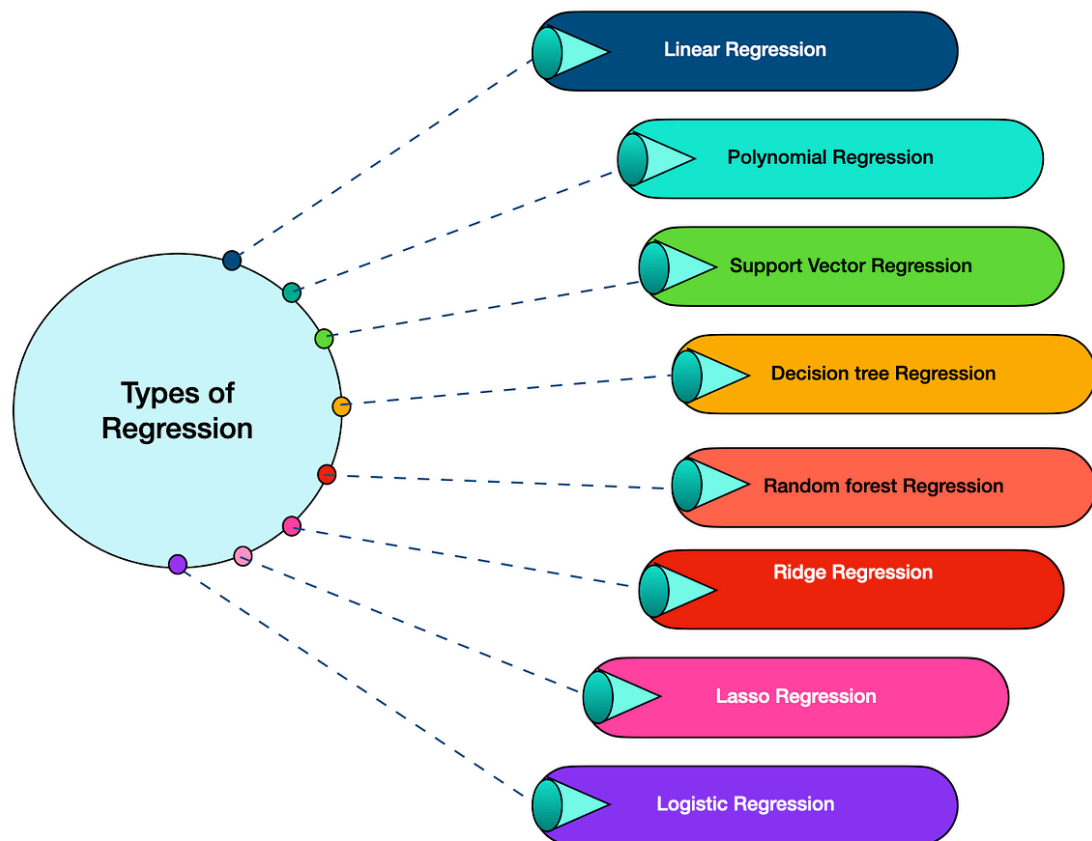
Regression is a type of **supervised learning** in **Machine Learning (ML)** used to predict a **continuous numerical value**.

In simple words:
If the output is a **number** (like price, salary, temperature), we use **regression**.

Example of Regression

Problem	Input	Output
House Price Prediction	Size of house, location, rooms	Price (₹ 50,00,000)
Salary Prediction	Years of experience	Salary (₹ 8,50,000)
Temperature Forecast	Date, humidity	Temperature (32°C)

Types of Regression in ML



1. Linear Regression

Used when the relationship between input and output is linear (straight line).

Example:

Predicting salary based on years of experience.

Formula: $y=mx+c$

- y = output(Predicted value)
- x = input
- m = slope
- c = intercept

Regression uses **Mean Squared Error (MSE)** to measure error. The goal is to **minimize MSE**.

$$MSE = \frac{1}{n} \sum (Y_{actual} - Y_{predicted})^2$$

2. Multiple Linear Regression

Used when there are **multiple features(input variables.)**

Equation: $Y = b_0 + b_1X_1 + b_2X_2 + b_3X_3 + \dots$

Example:

Predicting house price based on:

- Size
- Number of rooms
- Location
- Age of house

If: Size increases $\rightarrow$ price increases
 Age increases $\rightarrow$ price decreases

3. Polynomial Regression

Used when data is not linear but curved.

$$Y = a + bX + cX^2$$

Example:

Growth of bacteria over time (curved relationship)

Initially slow $\rightarrow$ then fast $\rightarrow$ then slows again.(This cannot be captured by a straight line.)

Growth prediction, Sales trends , Biological data

4. Ridge Regression (L2 Regularization)

Problem it solves:

Overfitting (model performs well on training but poorly on testing data).

Formula:

$$Cost = MSE + \lambda \sum w^2$$

It adds penalty on large coefficients.

Example:

Suppose we predict house price using 50 features.

Some features are not very important.

Ridge shrinks coefficients but does NOT make them zero.

5. Lasso Regression (L1 Regularization)

Formula:

$$Cost = MSE + \lambda \sum |w|$$

Special Feature: It can make some coefficients exactly zero → performs feature selection.

Example:

If predicting student marks using:

- Study hours
- Sleep hours
- Favorite color
- Shoe size

Lasso may remove "favorite color" and "shoe size".

Used when:

- Need automatic feature selection
- High-dimensional datasets

6. Elastic Net Regression

Combination of:

- Ridge (L2)
- Lasso (L1)

Formula:

$$Cost = MSE + \lambda_1 \sum |w| + \lambda_2 \sum w^2$$

Example:

Used when many correlated features exist (like medical data).

Balances shrinkage + feature selection.

7. Support Vector Regression (SVR)

Based on Support Vector Machine.

Key Idea:

Instead of minimizing total error, it tries to fit data within a margin (epsilon).

Example:

Stock price prediction where small errors are acceptable.

Used in:

- Complex patterns
- High-dimensional data

8. Decision Tree Regression

How it works:

Splits data into branches based on feature values.

Instead of line → creates tree structure.

Example:

Predict house price:

- If size > 2000 sq ft → high price
- If size < 1000 sq ft → low price

Advantage:

- Handles non-linear relationships
- Easy to interpret

9. Random Forest Regression

What it is:

Collection of many decision trees.

Each tree predicts → final output is average of all trees.

Example:

Predict sales using 100 trees → more accurate and stable.

Advantage:

- High accuracy
- Less overfitting than single tree

10. Bayesian Regression

Based on Probability:

Instead of fixed coefficients, it uses probability distributions.

Example:

Weather forecasting with uncertainty estimation.

Outputs:

- Prediction: 30°C
- Confidence: $\pm 2^\circ\text{C}$

Real-Life Example (Simple)

Imagine you run a coaching institute (like you might for ML students).

You want to predict:

- If a student studies 2 hours → marks = 40
- If a student studies 4 hours → marks = 70

You build a regression model to predict marks based on study hours.

Regression vs Classification

Regression	Classification
Predicts numbers	Predicts categories
Example: Price	Example: Pass/Fail

Probability for Machine Learning

In Machine Learning, **Probability** is the mathematical framework used to handle **uncertainty**. Since real-world data is rarely perfect (it has noise, missing values, and randomness), we use probability to quantify how likely a certain outcome or pattern is to be true.

Concepts in Probability

To understand how a machine "thinks" in chances, we use these three pillars:

- **Experiment:** Any process with an uncertain result (e.g., scanning an email to see if it's spam).
- **Outcome:** The specific result of one trial (e.g., the email *is* spam).
- **Random Variable (X):** A variable that takes on different values based on the outcome of a random event.
 - **Discrete:** Fixed values (e.g., predicting a "Yes" or "No" label).
 - **Continuous:** Range of values (e.g., predicting a house price of 25000,42000,50000).

Example:

The Spam Filter (Classification)

Imagine you are building a filter to catch "Spam" emails. The machine doesn't just say "This is Spam." Instead, it calculates the **Probability** based on words it sees.

You have a dataset where:

- **80%** of emails containing the word "Winner" are **Spam**.
- **5%** of emails containing the word "Winner" are **Not Spam**.

Comparison Table: Discrete vs. Continuous Probability

Feature	Discrete Probability	Continuous Probability
Data Type	Counts / Categories	Measurements / Scales
Example	Predicting "Pass" or "Fail"	Predicting the "Price" of a Stock
Visualization	Bar Chart / Histogram	Probability Density Function (Curve)
ML Use Case	Classification (k-NN, Naive Bayes)	Regression (Linear, Polynomial)

In Machine Learning, a **Random Variable** is the bridge between real-world uncertainty and mathematical calculation. It is not a "variable" in the traditional programming sense (like `x = 5`); instead, it is a function that maps the outcomes of a random process to numerical values.

A **Random Variable (X)** represents the possible outcomes of an uncertain event. Since ML models cannot process "maybe" or "perhaps," we use random variables to turn these possibilities into a **Probability Distribution**.

Types of Random Variables:

There are two main types, and the distinction is vital for choosing the right ML algorithm.

A. Discrete Random Variable

These have a countable number of distinct values. You use these for **Classification** tasks.

- **Example 1 (Coin Toss):** X can be 1 for Heads and 0 for Tails.
- **Example 2 (Email Filtering):** X can be 1 for "Spam" and 0 for "Not Spam."
- **ML Application:** Naive Bayes, Decision Trees.

B. Continuous Random Variable

These can take any value within a range or interval. You use these for **Regression** tasks.

- **Example 1 (House Price):** X could be any value like 250,340.50 or 250,340.51
- **Example 2 (Sensor Data):** The temperature of a server room recorded as 24.567°C.
- **ML Application:** Linear Regression, Support Vector Regression (SVR).

Example: Student's Result

Let's define a Random Variable X for a student's performance in the exam.

Outcome (Real World)	Random Variable (X)	Probability P(X)
Student Fails	0	0.05
Student Passes	1	0.65
Student Gets Honors	2	0.30

Key Properties:

1. **Numeric Mapping:** We converted "Fails/Passes" into numbers (0, 1, 2) so a computer can perform matrix multiplication on them.
2. **Total Probability:** The sum must always equal 01 ($0.05 + 0.65 + 0.30 = 1.0$).

Comparison Table

Feature	Discrete Random Variable	Continuous Random Variable
Values	Distinct/Isolated points.	Any value in an interval.
Summation	Uses sum (Sigma).	Uses int (Integration).
Example	Number of clicks on an ad.	Time spent on a webpage.

Statistics for Machine Learning

Statistics is the engine that allows us to make sense of data. While Probability deals with predicting future events, Statistics looks at historical data to find patterns, summarize information, and draw conclusions about a large group (population) based on a small subset (sample).

Statistics in ML

- **Data Cleaning:** Identifying outliers (wrong data points).
- **Feature Selection:** Deciding which data points actually affect the result.
- **Model Evaluation:** Measuring how "accurate" or "reliable" a prediction is.

The Two Main Types of Statistics : statistics is divided into two primary branches:

A. Descriptive Statistics

B. Inferential Statistics

A. Descriptive Statistics

This is used to **describe and summarize** the data you already have. It doesn't try to predict the future; it just tells you "what the data looks like."

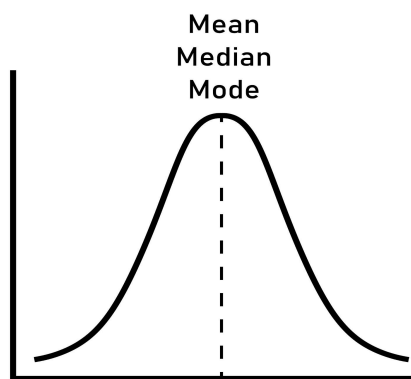
Key Tools:

- **Measures of Central Tendency:** Mean (Average), Median (Middle value), and Mode (Most frequent).
- **Measures of Dispersion (Spread):** Range, Variance, and Standard Deviation (how much the data varies from the average).

Example: You have the marks of 50 students in a class.

- **Mean:** The average marks of the class.
- **Standard Deviation:** Tells you if everyone got similar marks or if some got 90s while others got 20s.

Symmetric



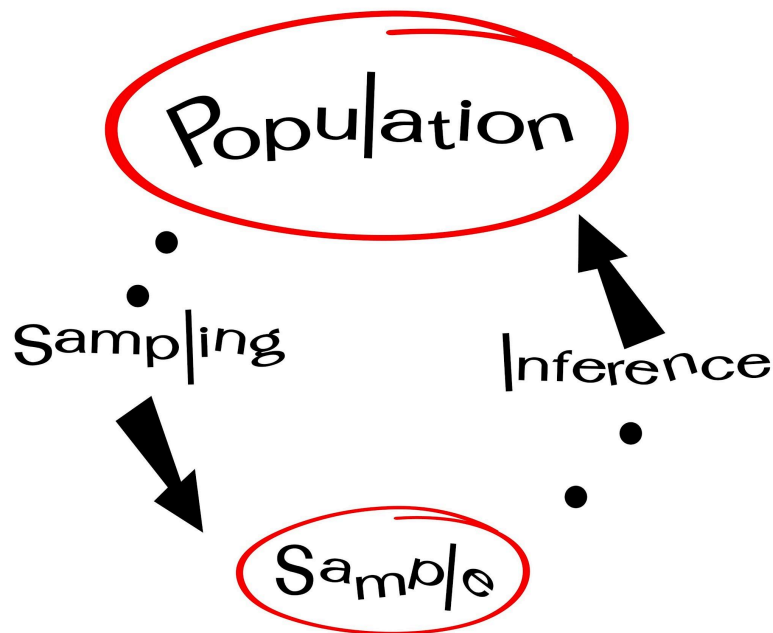
B. Inferential Statistics

This is used to **make predictions or inferences** about a large population based on a smaller sample of data. This is the heart of Machine Learning "training."

Key Tools:

- **Hypothesis Testing:** Testing if a pattern is real or just a coincidence (e.g., P-value).
- **Regression:** Predicting a numerical value based on trends.

Example: A drug company tests a new medicine on 1,000 people (Sample). They use **Inferential Statistics** to conclude that the medicine will be effective for millions of people (Population) across the country.



3. Comparison Table: Descriptive vs. Inferential

Feature	Descriptive Statistics	Inferential Statistics
Purpose	To summarize data features.	To make predictions/generalizations.
Data Used	The entire available dataset.	A small sample of the population.
Output	Charts, Tables, Averages.	Probabilities, Predictions, Tests.
ML Stage	Data Exploration (EDA).	Model Training and Testing.

5. Statistical Data Types

When you prepare your data for an ML model, you will categorize it into:

1. **Qualitative (Categorical):** Non-numeric data like "Color" or "Gender."
2. **Quantitative (Numerical):** Numeric data like "Height" or "Salary."
 - **Discrete:** Whole numbers (e.g., number of cars).
 - **Continuous:** Any value (e.g., weight, 65.4 kg).

Linear Algebra for Machine Learning

Linear Algebra as the "skeleton" of Machine Learning. While the algorithms (like Neural Networks or Regressions) are the brain, Linear Algebra provides the structure that allows them to hold and process massive amounts of information.

Linear Algebra is the math of "Lines and Grids." Imagine you are playing a video game where you can move a character left, right, up, or down. Linear Algebra is simply the language used to describe those movements and how they change if you grow bigger, turn around, or move faster.

The Three Simple Pillars

1. It's about "Lists of Instructions" (Vectors)

Instead of dealing with one number at a time, Linear Algebra deals with **lists**.

- **Simple Math:** $2 + 3 = 5$
- **Linear Algebra:** $[2, 4] + [3, 1] = [5, 5]$
- **Real-world:** Think of a GPS. It doesn't just give you one number; it gives you a **vector** (Latitude and Longitude) to tell you exactly where you are on a 2D grid.

2. It's about "Converting Tables" (Matrices)

A Matrix is just a box of numbers (like a small Excel spreadsheet). In Linear Algebra, we use these boxes as "filters" or "rules" to change our lists.

- **Example:** If you have a list representing a square's corners, and you multiply it by a "Rotation Matrix," the square tilts.
- **In ML:** The "Table" is the computer's memory, and the "Rules" are how it decides if a photo is a cat or a dog.

3. It's about "Finding the Path" (Systems of Equations)

If you have two lines on a piece of paper, Linear Algebra is the tool that tells you exactly where they cross.

- **In ML:** We have millions of "lines" (data points), and we use Linear Algebra to find the one "perfect line" that goes through the middle of all of them. This is how we make predictions.